

Yingrui Hou - Portfolio and Resume

Yingrui Hou - Portfolio and Resume

Clermont-Ferrand, France | yingrui.hou@outlook.com | +33(0)785396332 LinkedIn: <https://www.linkedin.com/in/yingrui-hou> | ORCID: <https://orcid.org/0000-0001-6454-278X> | Personal GitLab: <https://gitlab.com/hyrsophie> | CERN GitLab: <https://gitlab.cern.ch/yihou>

Positioning

Simulation & Data Science Engineer for Physical Systems

Best fit for industrial R&D teams working on physical systems, especially where simulation-to-measurement alignment, calibration, model validation, and uncertainty-aware analysis are central to the work. Strongest value comes when the goal is not generic analytics, but making models trustworthy enough to support engineering decisions.

Personal Introduction

From noisy measurements to reliable model decisions

I build calibrated models from noisy measurements and imperfect simulations for physical systems.

I build calibrated models from noisy measurements and imperfect simulations. My core strength is designing modelling, calibration, and validation strategies that extract reliable conclusions when the true state is only indirectly observed.

In postdoctoral work, that meant building reusable C++/Python/ROOT/Geant4 workflows for simulation-to-test alignment, calibration, monitoring, time-dependent inference, and model correction in large shared technical environments. The value came from turning validation results into calibration updates, software follow-up actions, and more credible modelling decisions.

This profile is strongest in industrial R&D environments working on complex physical systems, where instrumented tests, imperfect models, and engineering decisions must stay aligned under uncertainty.

Seeking

Simulation & Data Science Engineer for Physical Systems, with strongest fit in model validation, test-data analytics, and simulation-to-measurement alignment roles where calibrated models and robust technical decisions matter.

Overview

What this work supports in industrial R&D

Across these projects, the recurring pattern is consistent: build the workflow, correct mismatch between simulation and measurement, validate aggressively before trusting the result, and make the final model usable for decisions rather than just analysis. That is the same pattern behind industrial R&D teams working on physical systems, test-data interpretation, monitoring, calibration, model credibility, and simulation-informed design iteration.

Yingrui Hou - Portfolio and Resume

The strongest proof repeats across multiple cases: test data used to calibrate model response, structured bias removed through reusable correction logic, anomalies traced to instrument behaviour or software rather than left unexplained, and quantified performance changes used to support calibration choices, software fixes, and simulation updates before a result is trusted for engineering decisions.

Selected Outcomes

- Quantified response non-uniformity contribution to resolution below 1% in a detailed detector study
- Improved reconstructed-energy resolution by about 2x in representative PSD-based studies
- Built validation workflows that exposed detector and software issues, including a missing-energy bug later fixed downstream

Selected Work

Sensor Simulation & Reconstruction

Tags: Signal Processing, Simulation Calibration, Digital Twins

Built a simulation-and-measurement workflow to quantify how detector non-uniformity affects energy resolution, using beam-test data to constrain the model instead of assuming ideal sensor response.

Methods: Geant4 simulation, beam-test analysis, response-map reconstruction Impact: linked measured non-uniformity to system-level resolution through both simulation code and experimental validation. Industry relevance: Sensor Analytics | Measurement Systems | Simulation-Assisted Design

Pulse-Shape Discrimination for Energy Reconstruction

Tags: Pulse Shape Discrimination, Detector Simulation, Time-Resolved Signal Analysis, Resolution Optimisation

Built a PSD-oriented simulation and analysis workflow to extract shower composition from scintillation time profiles and use it to improve reconstructed energy.

Methods: particle-dependent timing models, component fitting, correlation-based correction Impact: showed that PSD-derived shower fractions can improve reconstructed energy and help judge which detector materials remain usable for timing-based separation. Industry relevance: Time-Resolved Sensing | Signal Decomposition | Measurement-System Design

Model Correction Workflow

Tags: Scientific Software, Event Filtering, Simulation Correction, Framework Integration

Developed simulation-correction tooling that turns measured physics constraints into an event-level resampling workflow.

Methods: Gaudi-package development, event-level correction, framework integration Impact: turned measured constraints into a reusable event-level correction package for improving simplified simulation realism. Industry relevance: Simulation Platforms | Scientific Computing | Model Correction

Yingrui Hou - Portfolio and Resume

Test-Data Validation & Root-Cause Analysis

Tags: Monitoring, Data Validation, Performance Diagnostics, Root-Cause Analysis

Built a production-style validation workflow for early Run 3 photon reconstruction, focused on finding actionable detector and software issues.

Methods: benchmark-channel checks, MC-data comparison, detector-level diagnostics Impact: translated validation discrepancies into concrete calibration, detector, and software follow-ups before they became accepted bias. Industry relevance: Production Monitoring | Model Validation | Observability | Quality Assurance

Time-Series Inference

Tags: Time-Series Modelling, Bias Correction, Calibration, Joint Fitting

Built a bias-corrected time-series inference workflow for data affected by selection effects, detector resolution, and sample mismatch.

Methods: acceptance modelling, resolution calibration, custom fit components, joint fitting

Impact: converted a bias-sensitive latent-dynamics problem into a controlled inference workflow with dataset-specific corrections. Industry relevance: Durability Modelling | State Estimation | Latent-Dynamics Inference

Domain Adaptation for ML

Tags: GBDT, CatBoost, Reweighting, Feature Validation

Built an ML validation and sample-alignment workflow in a shared analysis environment for noisy selection problems under dataset shift.

Methods: CatBoost training, train-test diagnostics, gradient-boosting reweighting Impact: improved downstream model reliability by treating dataset mismatch and overtraining checks as part of the production workflow. Industry relevance: Domain Adaptation | Applied ML | Dataset Shift

Additional Statistical Methods

These cases show deeper statistical-method development alongside the more immediately industry-facing simulation, calibration, and test-data workflows above.

Advanced Inference Workflow Validation

Tags: Advanced Statistical Modelling, Basis-Function Methods, Latent-Parameter Estimation, Collaborative R&D

Contributed the proof-of-principle validation needed to move a new basis-expansion inference idea from theory toward a usable workflow under realistic measurement conditions.

Methods: basis expansion, covariance-aware joint fitting, pseudoexperiment validation, statistical background subtraction Impact: helped move a novel inference method into a realistic workflow by testing robustness against background, efficiency, and resolution effects and by preparing purified real-data inputs. Industry relevance: Advanced Modelling | Inverse Problems | Signal Analytics | Collaborative Method Development

Weak-Signal Inference

Yingrui Hou - Portfolio and Resume

Tags: Mixture Models, Maximum Likelihood, Feature Selection, Statistical Validation

Built a reusable C++/Python/ROOT workflow for weak-signal extraction in data where background dominates and event-level truth is unavailable.

Methods: likelihood fitting, control-sample validation, production yield workflows Impact: turned rare-signal extraction into a reproducible workflow with stability checks and auditable validation logic. Industry relevance: Rare Event Analytics | Anomaly Detection | Weakly Labeled Systems

Resume

Summary

R&D-oriented simulation and data scientist with a background in experimental physics and several years of experience building reusable Python/C++ workflows for calibration, validation, and inference on noisy measurement systems.

Core strength: turning imperfect simulations and biased measurements into decision-ready models through simulation-to-data alignment, correction workflows, uncertainty handling, and robust validation.

Best fit: industrial R&D teams working on physical systems, instrumented tests, sensor data, model credibility, and simulation-informed engineering decisions.

Track record: delivered validation and correction workflows that fed calibration updates, software fixes, and quantified model improvements under uncertainty.

Selected Outcomes

- Quantified response non-uniformity contribution to resolution below 1% in a detailed detector study
- Improved reconstructed-energy resolution by about 2x in representative PSD-based studies
- Built validation workflows that exposed detector and software issues, including a missing-energy bug later fixed downstream

Skills

- Python
- C/C++
- ROOT / RooFit
- Geant4
- CatBoost / Gradient Boosting
- Probabilistic Modelling
- Statistical Inference
- Maximum Likelihood Estimation
- Model Validation
- Calibration & Bias Correction
- Simulation-to-Data Alignment
- Signal Processing

Yingrui Hou - Portfolio and Resume

- Domain Adaptation
- Workflow Automation (YAML / Shell / CMake)
- Large-Scale Data Analysis

Technical Highlights

- Model validation and uncertainty quantification for noisy measurement systems
- Simulation-to-measurement alignment through calibration, reweighting, and correction workflows
- Sensor and system modelling with Geant4, ROOT, Python, and C++
- Test-data analytics, anomaly tracing, and root-cause diagnosis
- ML validation under dataset shift for decision-facing pipelines

Professional Experience [Postdoctoral Researcher (2021 - 2025)]

Inference, Validation & Analysis Workflows

CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration

- Built reusable C++/Python/ROOT workflows for weak-signal extraction, time-dependent inference, and validation on noisy datasets where direct event-level truth was unavailable.
- Implemented likelihood-based models and joint fits to estimate latent parameters under sample mismatch, finite resolution, and biased observation conditions.
- Developed CatBoost-based selection and validation pipelines, including train/test diagnostics, scan studies, and reproducible model application.
- Aligned simulated and observed datasets through gradient-boosting reweighting and control-sample calibration before downstream inference and selection decisions.
- Contributed reusable fit components and uncertainty-stress checks for advanced weighted and unbinned inference workflows.
- Worked in large technical collaborations with shared codebases and review discipline, with validation outputs feeding calibration and model decisions.

Simulation, Calibration & Test-Data Interpretation

CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration

- Built and adapted Geant4/C++ simulation code for optical-response and detector studies, including geometry, stepping, and analysis outputs.
- Combined beam-test data, simulation, and response maps to model measurement-chain behaviour and quantify non-uniformity effects on system resolution, including a quantified constant-term contribution below 1%.
- Built time-resolved signal-analysis workflows that recovered component fractions from scintillation timing and improved reconstructed-energy resolution by about a factor of 2 in representative PSD studies.
- Developed ROOT/C++ calibration and correction workflows to reduce structured bias and compare performance before and after correction across operating conditions.

Yingrui Hou - Portfolio and Resume

- Built benchmark-based monitoring workflows to validate reconstruction and calibration quality, then traced anomalies to detector behaviour or software issues instead of stopping at descriptive analysis.
- Identified concrete issues that led to follow-up actions, including detector-specific anomalies and a missing-energy software problem later fixed in reconstruction software.
- Implemented event-level model-correction tooling inside collaboration frameworks to improve fast-model realism using measured constraints.

Education

PhD in Particle Physics

University of Chinese Academy of Sciences

BSc in Applied Physics

China University of Mining and Technology

Languages

- Chinese: Native speaker
- English: C1
- French: B1 to B2